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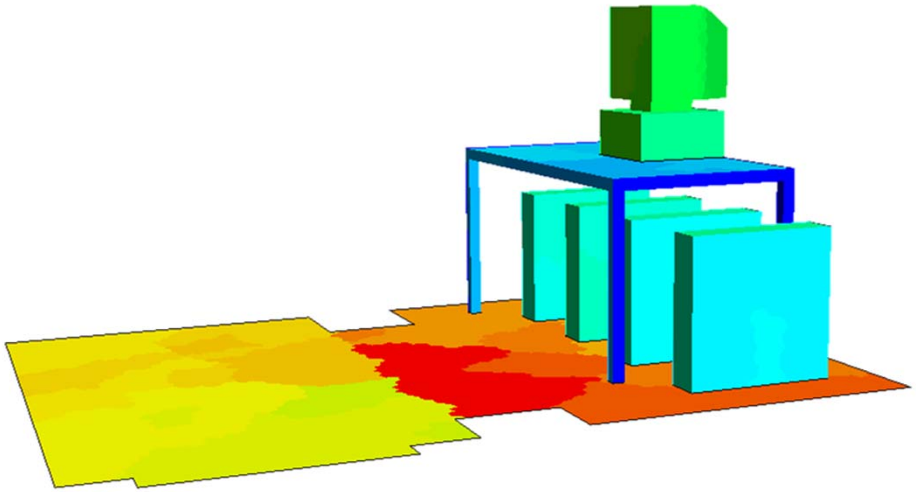
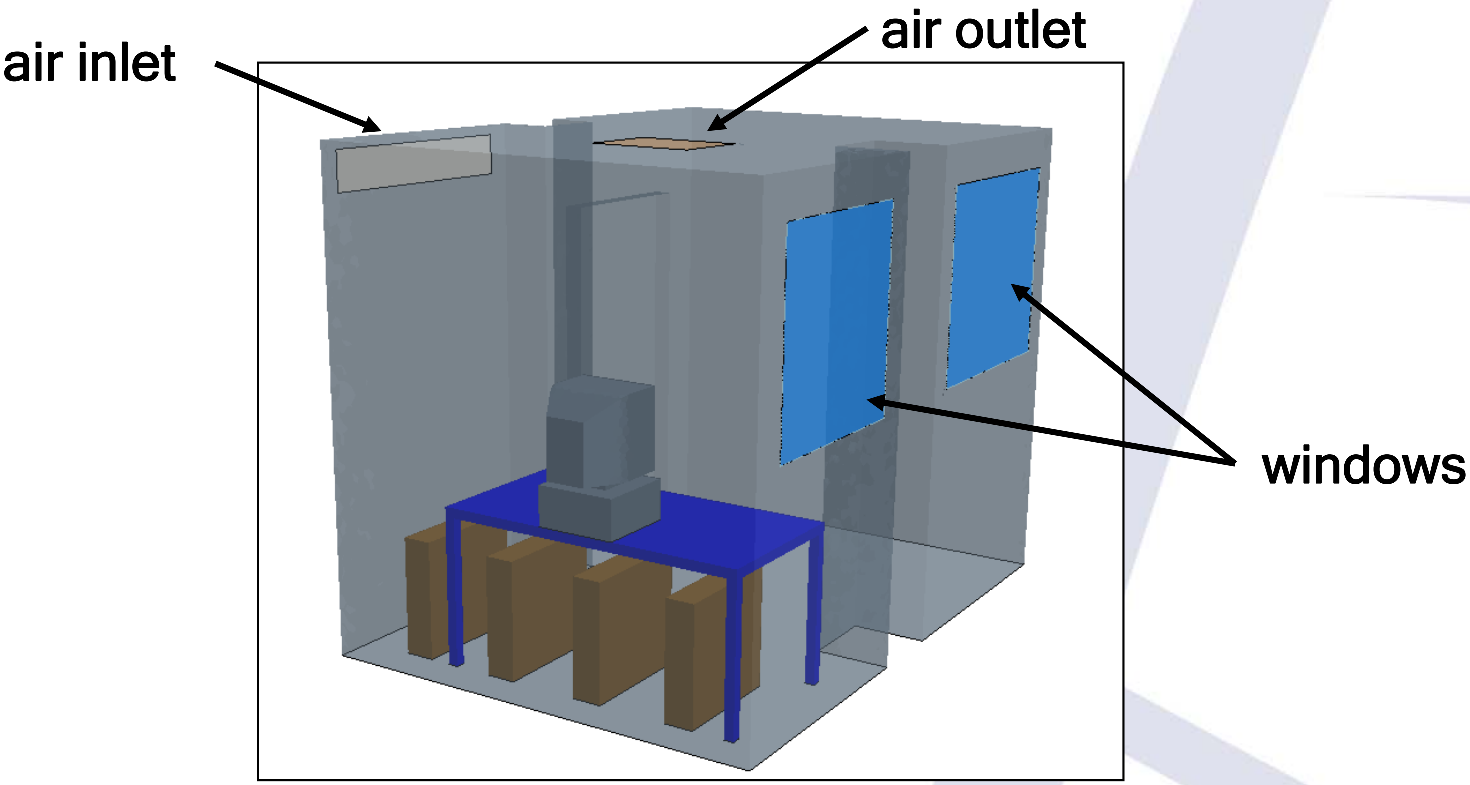


Workshop1 太阳辐射

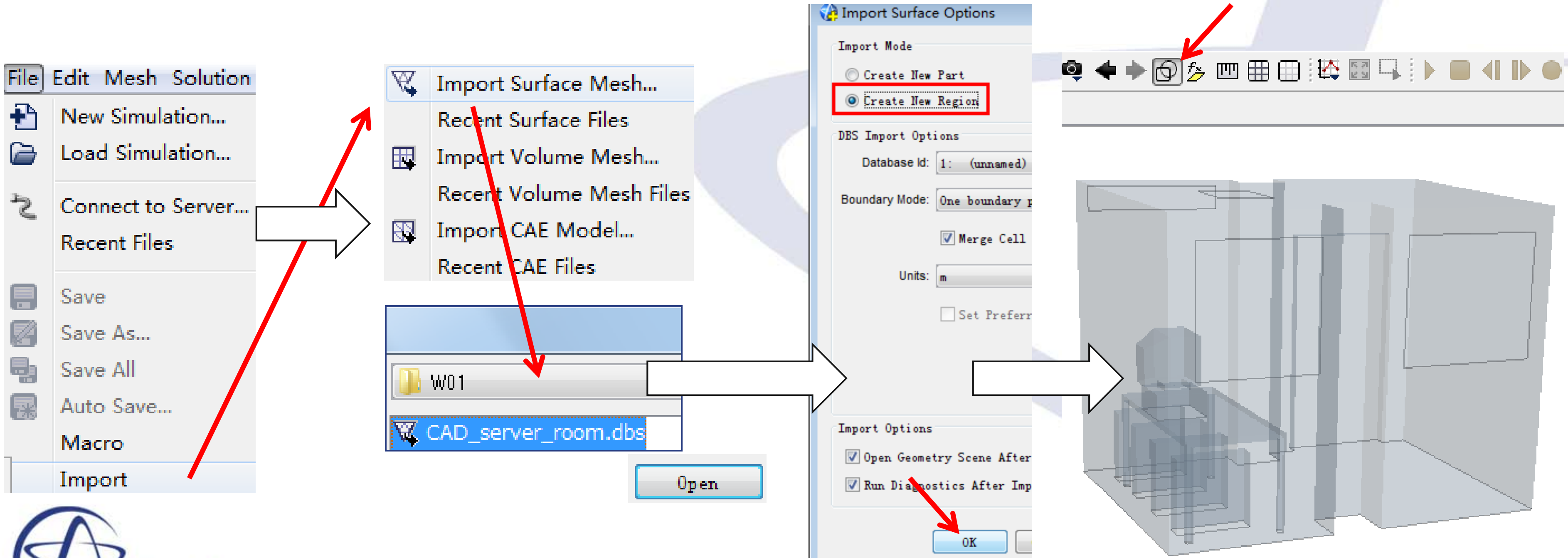
计算模型

- 太阳辐射经过房间的两个窗户。
  - 房间通过两个窗户换气。
  - 房间壁面绝热，不考虑重力。



导入几何

1. [File] > [New Simulation]
   新建一个模拟
2. [File] > [Import] > [Import Surface Mesh]
   选择CAD_server_room.dbs
3. 选择Create New Region,点击[OK]。



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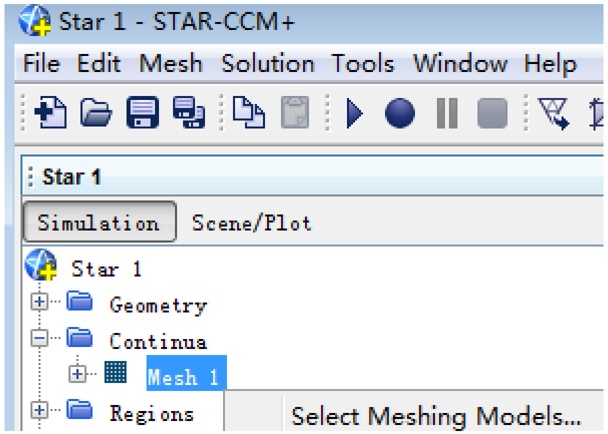
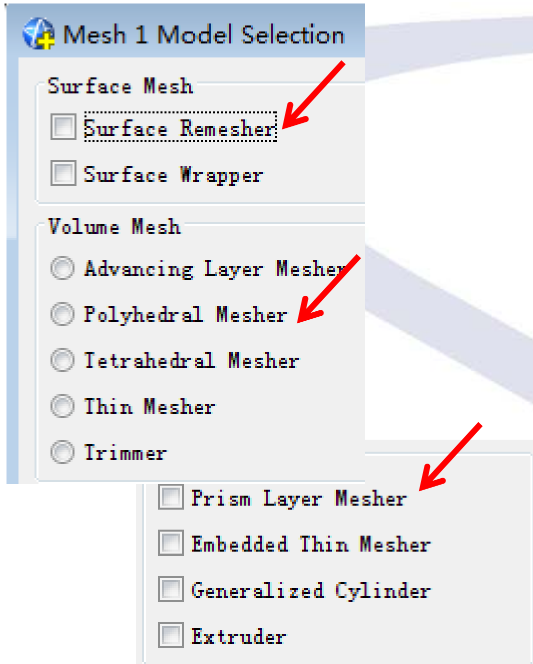
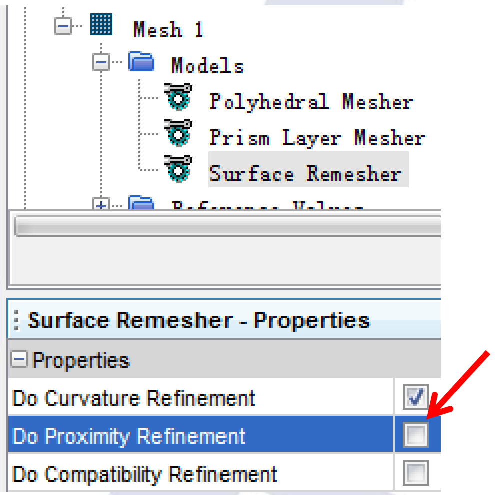
生成Surface Mesh和Volume Mesh

1. 右击[Continua] > [Mesh 1]， 选择[Select Meshing Models]
2. 选择Surface Remesher,Polyhedral Mesher和Prism Layer Mesher
3. [Surface Remesher]的Property中，将Do Proximity Refinement取消勾选

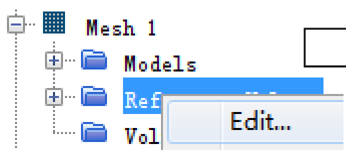


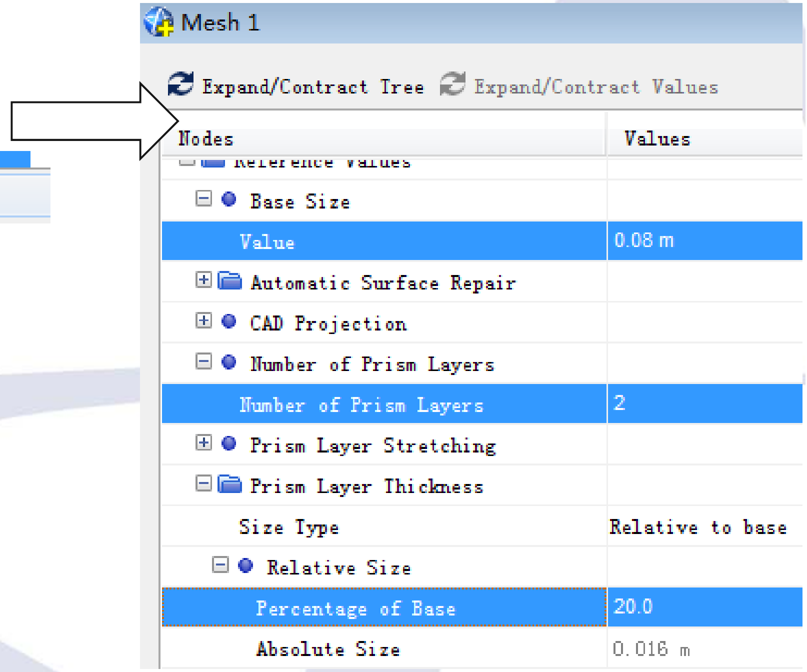


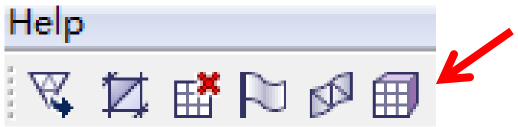


网格尺寸

1. 右击[Mesh 1]，选择[Edit]
   - Base Size=0.08 m
   - Number of Prism Layers=2
   - Prism Layer Thickness=20%
2. 点击 ，生成体网格



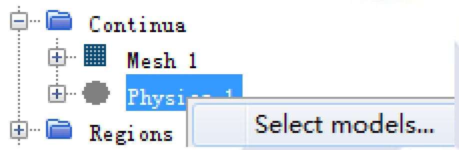


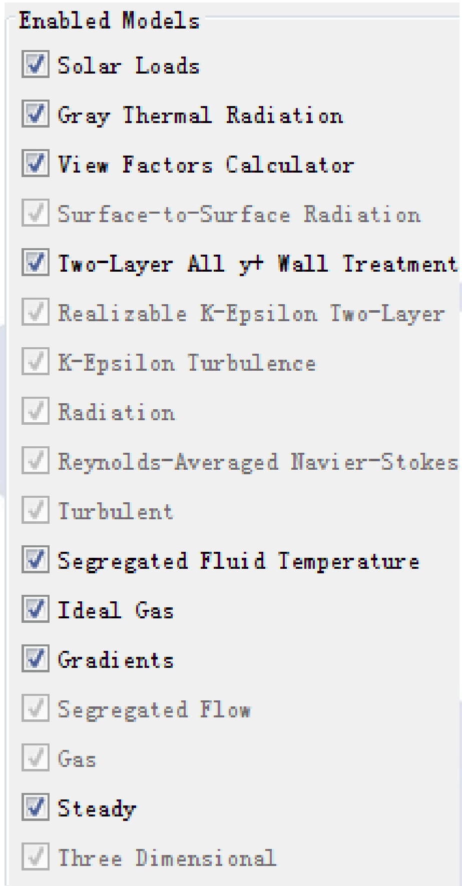


Volume Meshing Pipeline Completed: CPU Time: 19.73, Wall Time: 19.73, Memory: 93.26 MB
Cells: 59667 Faces: 328309 Vertices: 255608

物理连续体

1. 右击[Physics] ,选择 [Select models]
2. 依次选择下面的物理模型。
   - Steady
   - Gas
   - Segregated Flow
   - Ideal Gas
   - Turbulent
   - K-Epsilon Turbulence
   - Radiation
   - Surface-to-Surface Radiation
   - Gray Thermal Radiation
   - Solar Loads





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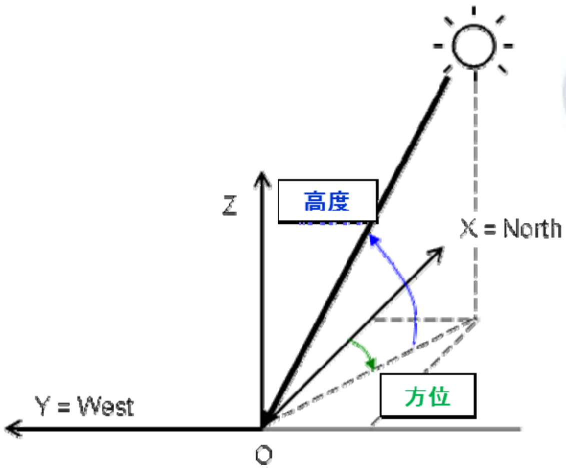
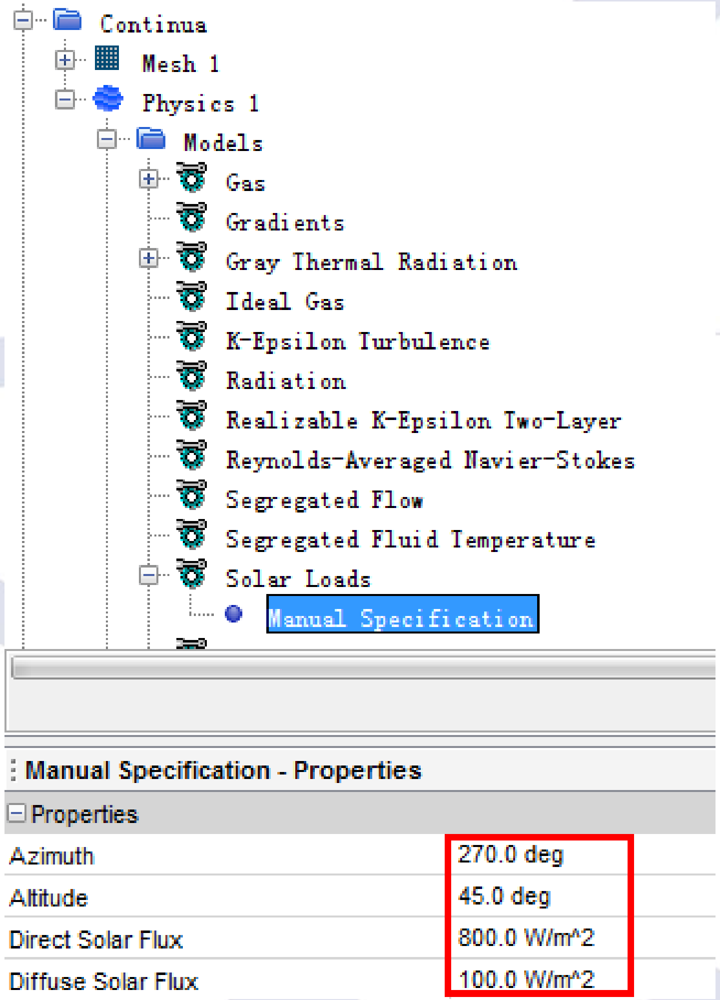
Solar Loads设定

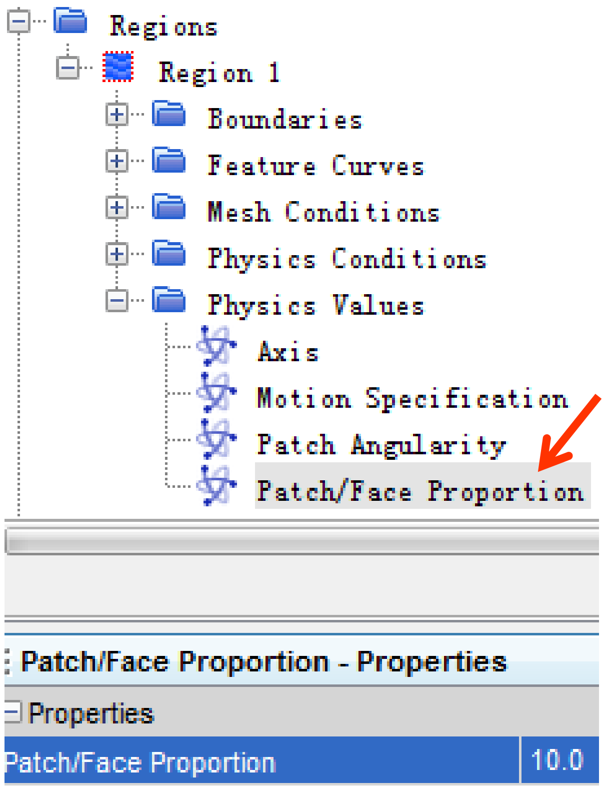
1. [Continua] > [Physics 1] > [Models] > [Solar Loads] > [Manual Specification],输入下面的值
   - Azimuth：270 deg（需要手动输入单位
   - Altitude：45 deg （需要手动输入单位）
   - Direct Solar Flux：800 W/m^2
   - Diffuse Solar Flux：100 W/m^2

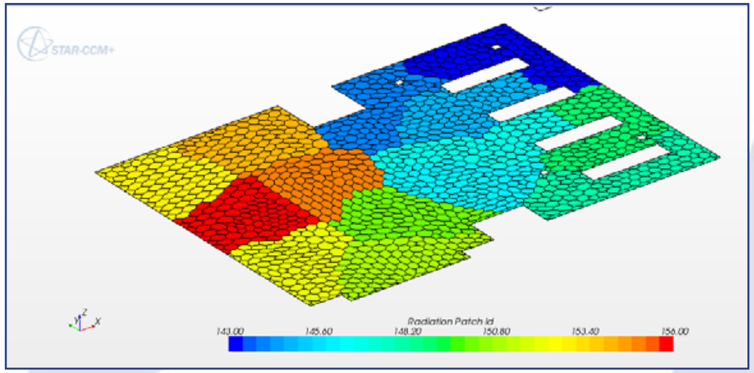




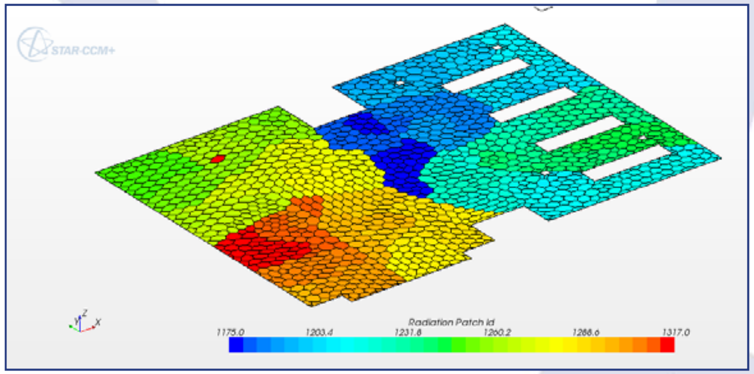
辐射设定

1. [Regions] > [Region1] > [Physics Values] > [Patch/Face Proportion],设置其值为10





Patch/Face Proportion设为1的时候



Patch/Face Proportion设为10的时候

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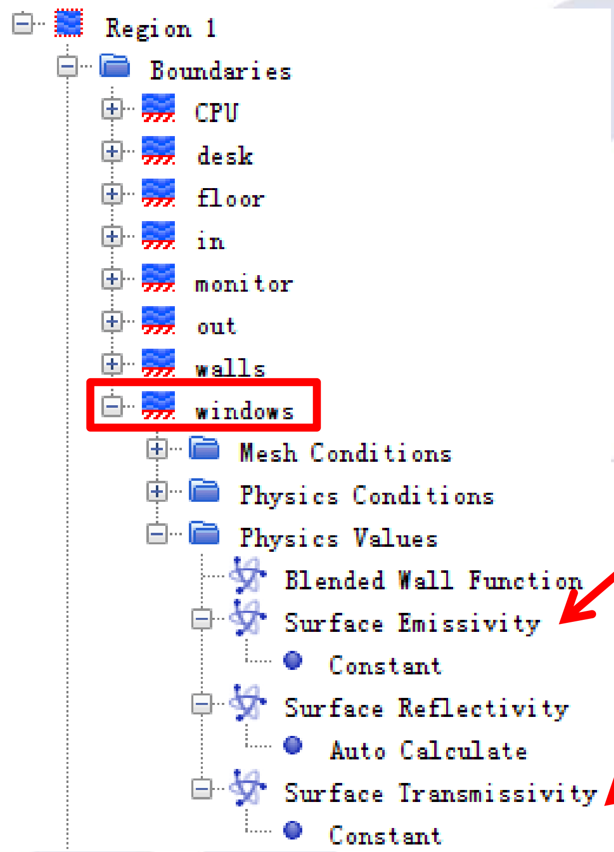
设定放射率

1. [Region] > [Region1] > [Windows] > [Physical Values]，设定表面放射率以及表面透射率
    - Surface Emissivity：0.1
    - Surface Transmissivity：0.4
        - Surface Reflectivity自动计算



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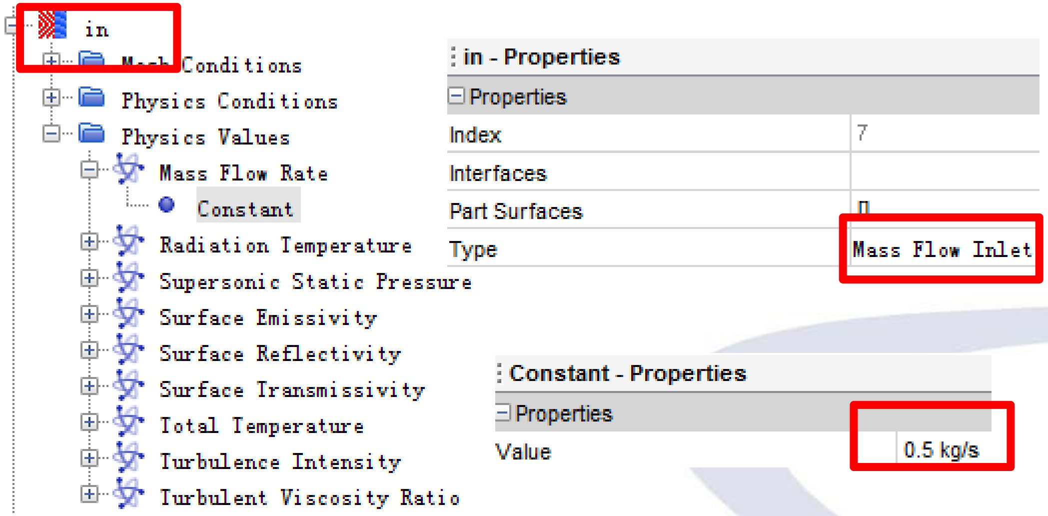
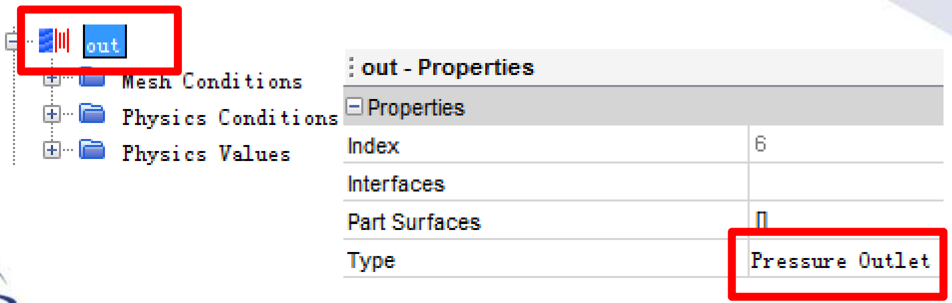
设定边界条件

1. [Region] > [Region1] > [in],将Type改为Mass Flow Inlet
   [Physics Value]中，设置质量流量为0.5kg/s



2. [Region] > [Region1] > [out],将Type改为Pressure Outlet



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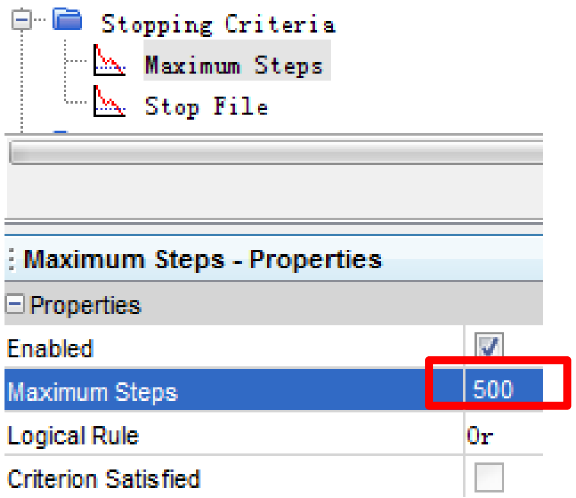
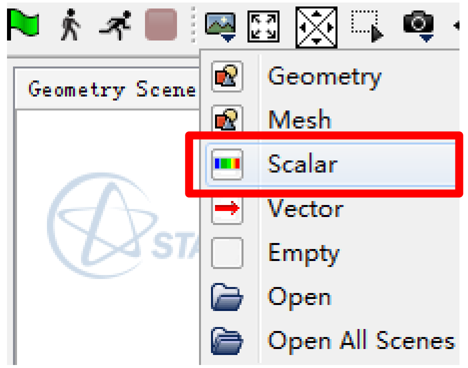
计算前的准备

1. [Stopping Criteria] > [Maximum Steps],设置为500。



2. 在计算开始前提前制作相关的标量云图。



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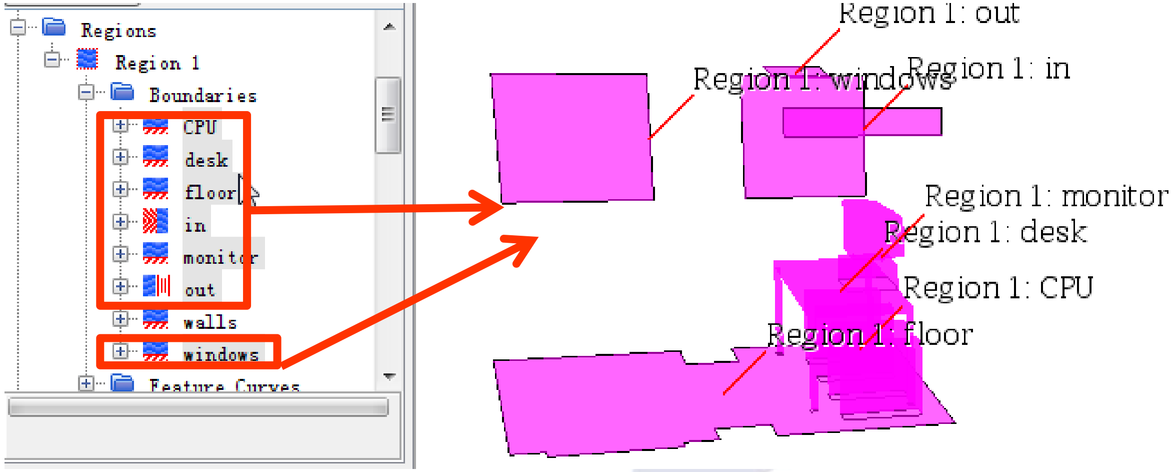
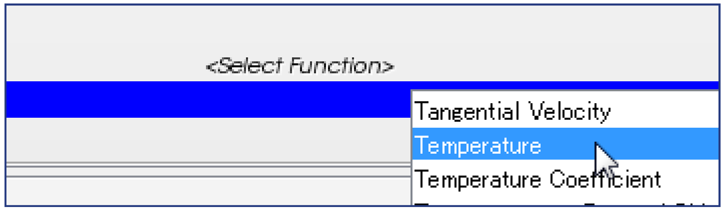
制作标量云图

3. 在[Region]中，选择除walls以外所有的边界。

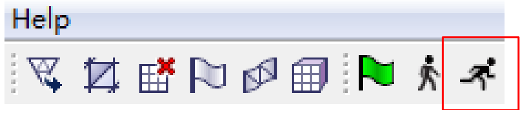


4. 右击<Select Function>，选择Temperature。

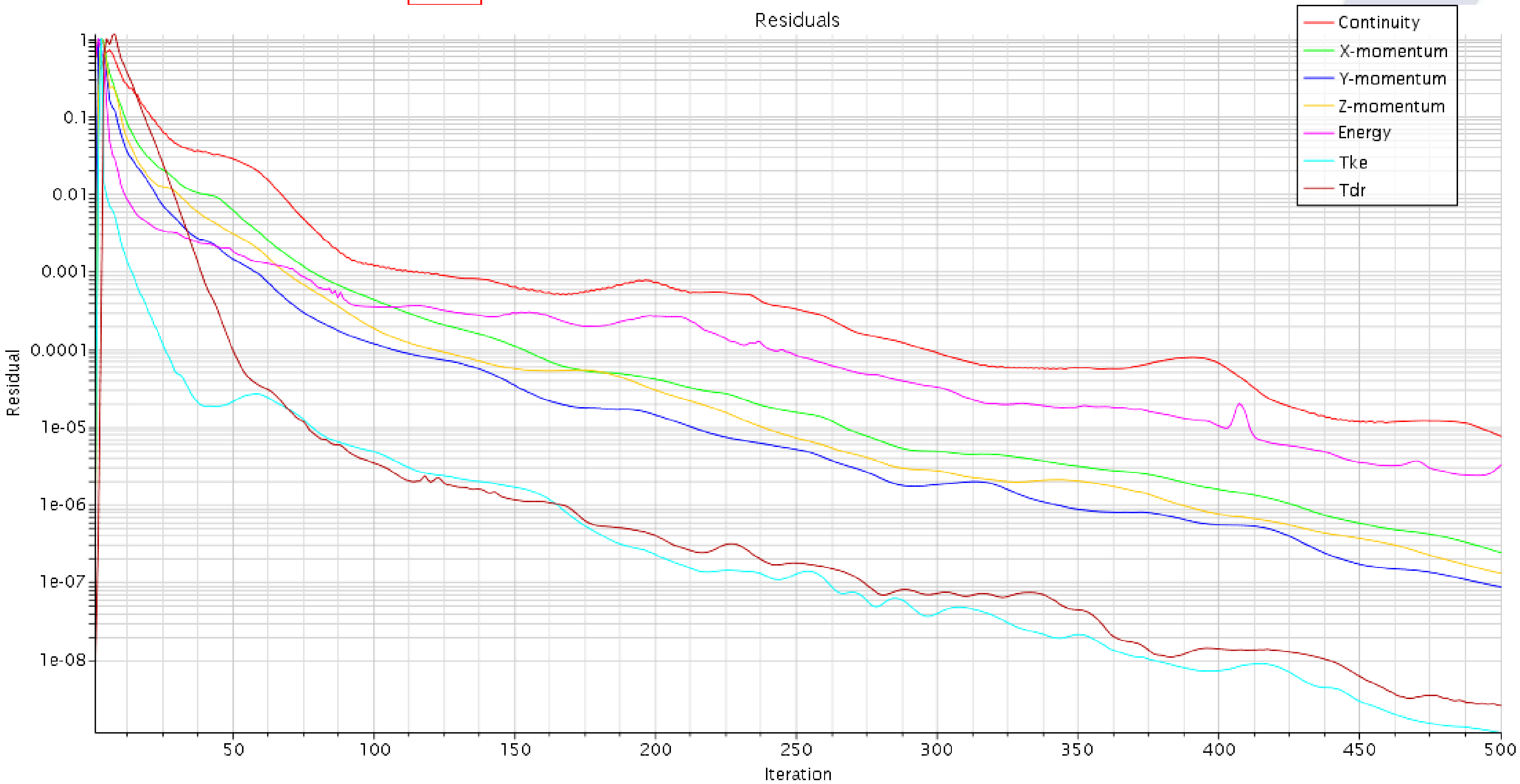


实行计算

1. 实行计算。



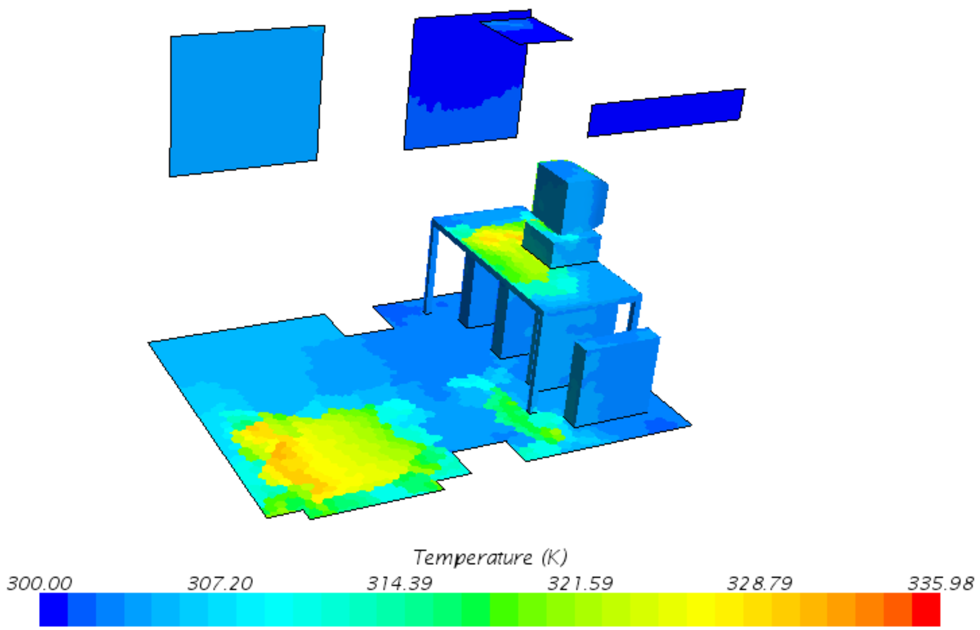
Help



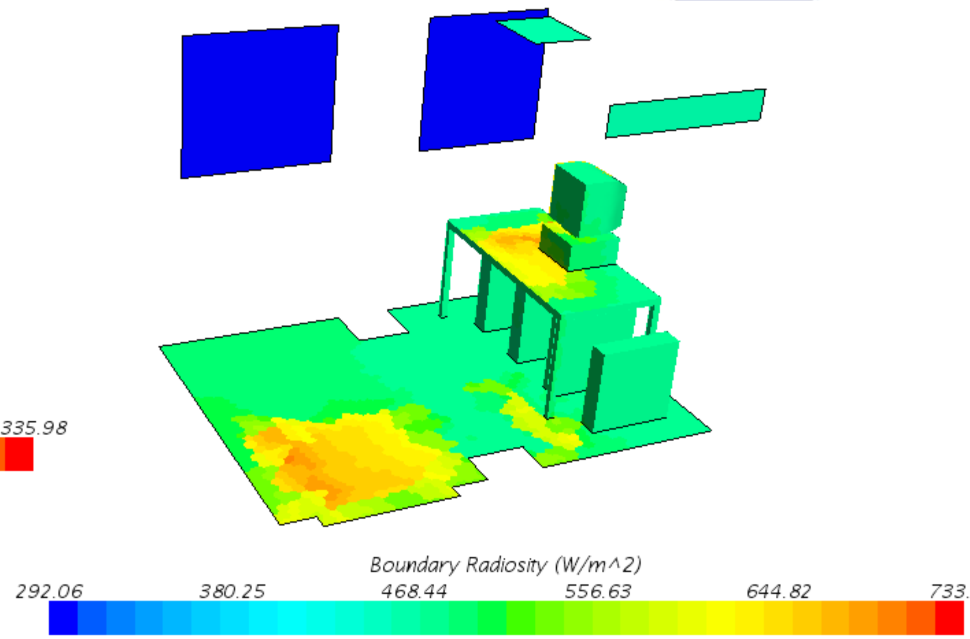
Residuals
Continuity
X-momentum
Y-momentum
Z-momentum
Energy
Tke
Tdr
Residual
1
0.1
0.01
0.001
0.0001
1e-05
1e-06
1e-07
1e-08
50
100
150
200
250
300
350
400
450
500
Iteration

后处理

1. 切换其他的标量确认计算结果。



Temparature



Boudanry Radiosity